	$\overline{R}$ = 0.37,	UCL = 0.78,	CL = 0.37,	LCL = 0
19.	$\overline{\overline{X}}$ = 30.1,	UCL = 44.75,	CL = 30.1,	LCL = 15.45
	$\overline{R}$ = 20.1,	UCL = 45.87,	CL = 20.1,	LCL = 0
20.	$\overline{\overline{X}}$ = 13.25,	UCL = 15.53,	CL = 13.25,	LCL = 10.97
	$\overline{R}$ = 3.12,	UCL = 7.12,	CL = 3.12,	LCL = 0
21.	$\overline{\overline{X}}$ = 41,	UCL = 43.31,	CL = 41.0,	LCL = 38.7
	$\overline{R}$ = 4,	UCL = 8.46,	CL = 4,	LCL = 0

Exercise 9.4

1	2	3	4	5	6	7	8	9	10	11	12	13
(d)	(b)	(d)	(d)	(c)	(a)	(c)	(b)	(b)	(b)	(a)	(d)	(c)
14	15	16	17	18	19	20	21	22	23	24	25	
(d)	(c)	(b)	(c)	(c)	(a)	(d)	(c)	(b)	(a)	(c)	(d)	

Miscellaneous problems

1. Three yearly moving Average
   148, 149.33, 152.33, 168.33, 253.33, 261.33, 281.67, 302.67, 327.
2. Four yearly moving Average
   708.75, 729.25, 748.25, 768.25, 784.5
3. Y = 55.975+ 0.825X
4. L= 49.9 P= 50.32 Fisher's = 50.09
5. Fisher =139.8
6. Consumer price index = 118.77
7. CLI = 126·10. The cost of living has increased upto 26.10% in 2011 as compared to 2010.
8. Control chart for Mean
   LCL = 47.56
   CL = 51.2
   UCL = 54.84
   Control chart for Range
   LCL = 0
   CL = 6.3
   UCL = 13.32
9. Control chart for Mean
   LCL = 1120.83
   CL = 1367.5
   UCL = 1614.17
   Control chart for Range
   LCL = 0
   CL = 427.5
   UCL = 904.16

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